Does the VIX Jump More When the S&P 500 Falls From a Calm Start?

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16 October 2026

ABSTRACT

Using 4,193 S&P 500 fall days from 1990 to 2025, I estimate the rise in the VIX per 1% fall with a least-squares model that compares days within the same calendar year. From the median prior VIX close (17.75), a 1% fall raises the VIX by about 1.4 points. This rise does not detectably depend on the prior VIX close, in 1990–2008 or in 2009–2025: between prior closes of 12 and 30, the data allow a change in the rise of at most about 0.2 points in 1990–2008 and 0.5 points in 2009–2025. A risk manager revaluing an option book can therefore apply a fixed number of VIX points per 1% fall.

1. Hypotheses

A trader or risk manager who wants to know what an option book is worth after a market fall must assume how far implied volatility will move. One way is to assume a fixed number of VIX points for each 1% fall. That is only safe if the number does not change with market conditions.

The first condition to check is the starting level. A 1% fall is a larger surprise when the market is calm than when it is already nervous, so calm starts may jump more. This is the intuition behind the project. It gives the first question.

Q1. Does the rise in the VIX for a 1% fall depend on the previous day's VIX close, when days within the same calendar year are compared?

> **H_{01}:** In each period (1990–2008 and 2009–2025), the rise in the VIX for a 1% fall does not depend on the previous day's VIX close.
> **H_{11}:** In each period, the rise depends on the previous day's VIX close; we expect a larger rise when the previous close is low.

The second condition is time. The sample covers 36 years, including the 2008 financial crisis and the 2020 pandemic. A relationship that holds in one period and not in another is a poor basis for revaluation. It gives the second question.

Q2. Is that dependence the same in 1990–2008 as in 2009–2025?

> **H_{02}:** The dependence is the same in both periods.
> **H_{12}:** The dependence differs between the periods.

2. Data

2.1 Data gathering

Market	Type	Content	Source	Coverage
U.S. equities	S&P 500 price index	Daily closing level	GitHub mirror vijinho/sp500 to 2016-10-20, then FRED series SP500	1990-01-02 to 2025-12-31
U.S. equity options	Cboe Volatility Index (VIX)	Daily closing level of the volatility implied by S&P 500 option prices	Cboe VIX_History.csv	1990-01-02 to 2025-12-31

2.2 Descriptive review

- **Window.** 2 January 1990 to 31 December 2025 (36 calendar years), split at 2009 into 1990–2008 and 2009–2025. The 32 dates with a close in only one series are dropped.

- **Fall days.** A fall day is a day on which the S&P 500 closes strictly below the previous close. There are 4,193 (46.3% of 9,062 daily changes): 2,258 in 1990–2008 and 1,935 in 2009–2025, and 96 to 143 per year.

3. Literature review

This project relies on three studies. Hibbert, Daigler and Dupoyet (2008) show that daily VIX changes are asymmetric in the S&P 500 return: from 1998 to 2006, falls raise the VIX more than rises lower it, and the relation fits best on the largest returns. Sarwar (2012) reports for 1992–2011 that the VIX–return relation "is much stronger when VIX is both high and more volatile". Giot (2002, working paper) estimates the relation between implied volatility and returns for the NASDAQ 100 separately in a bull market (1995–2000) and a bear market (2000–2002): it is negative in both, and implied volatility rises more after falls than after rises only in the bull market.

Hibbert et al. are the premise for the model, which estimates the rise on fall days and includes the fall size and its curvature. Sarwar is the premise for Q1: the response depends on the VIX level, in the direction opposite to H_{11}, which is therefore tested two-sided. Giot is the premise for Q2: the response can differ between market periods, so the dependence on the previous day's VIX close may differ between 1990–2008 and 2009–2025.

4. Methodology

4.1 Model

Equation (1) measures how the VIX change on a fall day depends on the fall size and the start level.

$$\Delta V_t = \alpha_y + \gamma_y \cdot x_t + \psi \cdot x_t^2 + [\ \beta_y + \delta \cdot x_t + \theta \cdot P_y \cdot x_t\ ] \cdot m_t + \kappa \cdot m_t(m_t - 1) + u_t \qquad (1)$$

4.2 Variables

Part	Symbol	Meaning
VIX change	ΔV_t	VIX close on fall day t minus the previous trading day's close, in VIX points
Start level	x_t	previous day's VIX close minus its average over the fall days of year y
	α_y	VIX change at the year's average start level, before any fall effect; separate for each year y
	γ_y	effect of x_t on the VIX change; separate for each year y
	ψ	effect of x_t squared on the VIX change
Fall size	m_t	$-100 \cdot \ln(S_t / S_{t-1})$ in percent, with S_t the S&P 500 close
	β_y	VIX rise per 1% fall at the year's average start level; separate for each year y
	δ	change in that rise per +1 VIX point of start level in 1990–2008 (tested in Q1)
	P_y	0 for 1990–2008, 1 for 2009–2025
	θ	additional change in that rise per +1 VIX point in 2009–2025 (tested in Q2)
	κ	effect of the squared fall size, written $m_t(m_t - 1)$ so that β_y is the rise at exactly 1%
Error	u_t	error term

4.3 Test procedure

1. **Sample.** The 4,193 days from 1990 to 2025 on which the S&P 500 closed below the previous day's close, split into 1990–2008 and 2009–2025 to compare the two periods. The split at 2009 was fixed before estimation as the one that compares them most precisely.
2. **Estimation.** The model in 4.1 is fitted by ordinary least squares, which picks the coefficients that make the squared gaps between fitted and actual VIX changes smallest. Each year has its own level, so only days of the same year are compared. The standard errors of least squares treat each day's unexplained VIX change as unrelated to the other days', but days of the same year are likely related, which makes those standard errors too small. So artificial samples with no true effect are built by

flipping the sign of all errors of a year together using a wild cluster bootstrap-t (Cameron, Gelbach and Miller, 2007), which keeps the errors of a year related as in the data.

3. **Outcome.** δ, the change in the rise per +1 VIX point of the previous close, decides Q1 for 1990–2008; $\delta + \theta$ decides it for 2009–2025, where θ is the extra change in 2009–2025. θ decides Q2. A null hypothesis is rejected when its p-value is below 0.05.
4. **Controls.** Two controls keep δ from picking up other effects. The previous close: it moves the VIX change even without a fall and is correlated with the fall size within years, so γ_y and ψ, its effect and the effect of its square, remove this. The fall size squared: the rise may not grow in step with the fall (Hibbert et al.), so κ, its effect, removes this.

5. Results

5.1 Main estimates

Test	Estimate	p-value	Decision at 5%
Q1, 1990–2008 (δ)	−0.00605	0.130	H_{01} not rejected
Q1, 2009–2025 ($\delta + \theta$)	−0.01258	0.339	H_{01} not rejected
Q2 (θ)	−0.00653	0.615	H_{02} not rejected

Estimates are in VIX points of rise per 1% fall, per +1 VIX point of the previous close. No null hypothesis is rejected at 5%: the rise does not detectably depend on the previous close in either period, and the periods do not detectably differ. The negative sign is the direction expected in H_{11}. For a 1% fall from the median previous close (17.75), the fitted rise, averaged over fall days, is 1.36 points. Between previous closes of 12 and 30, the 95% intervals allow the rise to change by −0.21 to 0.00 points in 1990–2008 and by −0.52 to +0.07 points in 2009–2025. The interval for 1990–2008 reaches 0.00 while p = 0.130 because the p-value assumes no effect in the resamples and the interval does not.

5.2 Robustness

δ and θ are re-estimated in nine alternative versions of the model: controls removed or changed, years left out, a shorter window and two-day changes. The decision matches 5.1 in 17 of 18 tests. The exception is 1990–2008 without ψ, where p = 0.037.

6. Interpretation

Main result. The data do not support that the VIX jumps more when the market falls from calm. For a 1% fall, no difference in the rise between calm and nervous starts can be detected in either period, and none between the two periods. The sign of the estimate is the expected one, a larger rise from a calm start, but it is too small to tell apart from zero. Changing the model does not alter this: in 17 of 18 checks, the answers to Q1 for 1990–2008 and to Q2 stay the same. The one exception is the model without ψ (the effect of the previous close squared), where a difference appears in 1990–2008, because without ψ the model credits to the fall an effect that the previous close has on the VIX anyway.

Implication for revaluing an option book. A trader or risk manager revaluing an option book after a market fall can assume a fixed VIX rise per 1% fall and need not enlarge it for a calm start, since the data show no such dependence in either period. This removes the need to forecast how much a calm market would overreact.

Limits

- The return and the VIX move are determined together, so the result shows co-movement, not cause. A calm start and a larger surprise cannot be separated.
- The period comparison rests mainly on 2008, 2009 and 2020, so a difference between periods is hard to detect with so few crisis years.
- The level of the rise differs by period: the yearly rise averages about 1.0 in 1990–2008 and about 1.75 in 2009–2025 (simple averages, not tested). A single fixed number does not fit both, which opens the question of why the level changed.

7. Sources

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